I thank the WRC program for the opportunities this Chair has afforded over the past four years. I arrived at Western in April 2022 with three goals: to support Western's bioinformatics initiatives, to integrate Western with national and international networks, and to advance the three core directions of my own research platform.

1. Computational infrastructure for biomedical data science

The WRC enabled me to lead a successful CFI John Evans Leaders Fund (JELF) application with co-applicants Guthrie (co-PI, M&I), Hu (co-PI, Biochem), Andrews (Biochem), Castellani (PLM), Dumeaux (ACB), Shooshtari (PLM), Gloor (Biochem), and Poon (PLM). The nine investigators have substantial computational components in their research but conflicting requirements (security and privacy controls, GPU capacity, network throughput, sensitive-dataset hosting) that the Digital Research Alliance of Canada cannot meet because of clinical or industry restrictions.

Drawing on my experience building the computational facility at McGill's Goodman Cancer Centre (a facility comparable in scope to the forthcoming Bioconvergence Centre), I designed a flexible distributed network rather than a single large server: GPU and multi-core nodes, large-capacity storage with local backup, a web host, a hardened machine for sensitive patient and industrial data, and personal workstations. The modular Dell-based architecture supports incremental addition and retirement of components, so the system remains current under continual micro-investment rather than periodic wholesale replacement. We also contributed a sizeable server to SHARCnet, giving members priority access to the national platform.

The system now serves approximately 80 active researchers across 16 groups (roughly 30 concurrent users daily; 120 unique users over three years), and offers Western a cost-effective path for onboarding new faculty: incoming hires gain immediate access to existing facilities and contribute from start-up or CFI funds only what is needed to keep the platform current. When NIH tightened policy on sensitive-dataset hosting in January 2026 (requirements for self-encrypting drives, WTS-secured facilities, monitored network access) our `biodatstage` server was readily adapted and now serves as the temporary solution for all Western researchers handling NIH-protected data. Installation and ongoing administration have benefited from sustained help from WTS, Schulich IT, the Dumeaux lab, and Greg Gloor.

2. Bioconvergence

I serve on the Schulich standing committee for research IT and on the Bioconvergence Centre IT planning committee; our CFI JELF experience has impacted decision making on both committees. Moreover, WTS has assigned dedicated personnel (Ed Gibson, Jason Oliver) to research IT, and there is now a clearer institutional understanding of the data volumes and computing modes that Bioconvergence members will require.

3. Artificial intelligence in research and teaching

Large language models and agentic tools (ChatGPT, Claude, Claude Code) are, in my view, the most consequential scientific advances I have witnessed in my career; the pace of development continues to accelerate. These tools will reshape research, teaching, and administration. Having worked with machine learning, statistical inference, and AI throughout my career, I am well positioned to help Schulich (i) develop best practices for AI in biomedical research and teaching, (ii) build AI tools for our scientists, and (iii) track AI-driven teaching innovations. Items (i) and (ii) are the subject of the accompanying WRC research proposal. Toward (iii), I delivered a Department of Biochemistry seminar demonstrating how agentic AI can substantially assist day-to-day research even for users with limited IT background (seminar materials online), and I have built AI components into both undergraduate and graduate courses (below).

4. National and international positioning

I serve as a Western ambassador to the Temerty Centre for AI Research and Education in Medicine (T-CAIREM), and I contribute the computational AI component to D. Litchfield's NSERC CREATE biotherapeutics proposal. To raise Western's visibility to prospective graduate students, a particularly timely need following the recent biomedical-data-science cluster hire, I organized a full-day open house featuring 24 short faculty research presentations, attended by 39 Western undergraduates and 11 external students via Zoom. I have also worked to give early-career researchers visibility: Andrews, Dumeaux, and Shooshtari served on the scientific committee for my annual Systems Biology workshop in Barbados. The full impact is difficult to assess at this stage, but the channels are now in place.

5. Teaching: bioinformatics, data science, and AI

BIOCHEM 9552. A graduate course I designed and teach in the Department of Biochemistry, meeting weekly to discuss quantitative methods, tools, and techniques for biomedical research. Students join from varied backgrounds; the course adapts to bring each student's skills to a level appropriate for thesis work, with individual emphasis on statistics, scientific software, programming, or AI as needed.

MEDSCI 3391. I co-designed and contribute one of four modules to Western's new undergraduate bioinformatics course, covering the foundations of machine learning and AI.

RESEARCH EXCELLENCE

My research program consists of three integrated lines of work, each combining wet-lab experimentation (basic molecular biology, genomics including DIY single-cell sequencing, and microscopy) with computational work that ranges from building novel statistical-inference, machine-learning, and deep-learning tools to the sound use of existing ones. Such tightly integrated wet-and-dry labs remain relatively rare; in my view this integration of design, experiment, and analysis is critical for trainees at every level. Fuller descriptions of my research interests and a software inventory are at mikehallett.science.

Combinatorial intervention sequencing (COIN-seq)

COIN-seq is a novel high-throughput functional genomics assay in which CRISPR-based perturbations pseudo-randomly knock-out or over-express several genes concomitantly in a pool of cells. The perturbations are delivered by lentiviral particles so that each cell receives a random subset of the chosen targets, allowing us to explore the space of viable genetic interactions and infer the underlying gene-regulatory relationships. Like its PERTURB-seq predecessors, COIN-seq uses single-cell transcriptomics to demultiplex the resulting pool of differentially perturbed cells; the informatics challenge starts with constructing robust transcriptional profiles for each colony.

The first COIN-seq publications are in preparation, based on data we obtained applying the platform to genes regulating the Epithelial-Mesenchymal Transition (EMT). Two graduate students, one research associate, and 12 undergraduates have completed thesis work on COIN-seq at Western.

Funding. My NSERC Discovery was renewed in the first term of this WRC and is now the primary support for the project (Total $264K$; CRC Tier 1 funds at Concordia originally launched it, and a MITACS award supports one student. We will submit to the CIHR Genomics (GMX) panel in Fall 2026 with a focus on partial EMT in early breast lesions.

Phenotypic and morphological heterogeneity in *Candida albicans*

During the previous term of this WRC we completed our first effort to identify phenotypic heterogeneity in the human fungal pathogen *C. albicans*. Phenotypic heterogeneity refers to the situation in which two isogenic organisms behave differently in the same environment, which we show occurs during the development of drug tolerance. The work used a novel single-cell transcriptomics assay we built by modifying the DROP-seq protocol; I published it as last author with J. Berman (Tel Aviv) and M. Whiteway (Concordia) in *eLife* (Dumeaux *et al.*, 2023). A follow-up collaboration is underway with L. Xie (Stanford).

To complement the transcriptional data with imaging, I built a suite of deep-learning tools called *Candescence*. The original effort (Bettauer *et al.*, *Microbiology Spectrum* 2022) was repurposed for several follow-up projects, including a collaboration with the L. Cowen lab (Toronto) examining intraphagosomal filamentation and macrophage escape (Case *et al.*, *mBio* 2023). My MSc student H. Zhang has extended Candescence to study colony growth phenotypes; a manuscript in collaboration with J. Berman is nearing completion.

Funding. Candescence was originally supported by my CRC Tier 1 at Concordia and is now sustained by NSERC and start-up funds at Western. Proposals are in preparation for ERC Synergy funding with J. Berman and a team at Charité, Berlin.

Breast cancer genomics and informatics

This is my longest-running line of work (since 2007), spanning both the wet-lab generation of molecular data and the computational methods to interpret it. It has three subcomponents.

Classification, prognosis, and prediction. My group has published several high-impact manuscripts on intrinsic subtypes and on prognostic and predictive gene signatures (the latter for therapeutic benefit), including Finak *et al.* (*Nature Medicine* 2009) and Paquet & Hallett (*J. Natl. Cancer Inst.* 2015). This remains a major aim, with current collaborations including B. E. Hildreth (Alabama).

Tumour, microenvironment, and systemic response. Our manuscript Tofigh *et al.* (*Cell Reports* 2014) showed that patients stratify into groups whose outcome is easy to predict from molecular features and groups where prediction is inherently difficult, introducing the concept of *inherent prognostic difficulty*. We see this as some of the first quantitative proof that microenvironmental and systemic components are the primary determinants of tumoral progression in some individuals. To investigate this further with the Dumeaux lab, we developed the MIxT framework for principled comparison of transcriptional programs across matched tissues (Dumeaux *et al.*, *PLOS Comput. Biol.* 2017; see MIxT) and demonstrated that peripheral-blood gene expression carries prognostic information independent of and complementary to the primary tumour. Understanding how the tumour and its systemic context shape one another remains central to our work. This work continues with the Dumeaux lab and my research associate E. Mucaki. For example, we are finalizing a manuscript that examines how specific microenvironmental processes interact with the epithelial tumor component in non-invasive early breast lesions.

The invasive switch (DCIS to invasive ductal carcinoma). DCIS is a non-obligate precursor to invasive disease, but current biomarkers cannot distinguish indolent from progressive DCIS at time of diagnosis. Because of the difficulty we encountered profiling archival FFPE tissue, which is the standard storage form for early breast lesions, we recently developed PREFFECT, a deep generative model (variational autoencoder with iterative refinement) that imputes high-quality transcriptomic profiles from FFPE material (Mucaki *et al.*, *J. Translational Medicine* 2025). Across two CIHR Project awards over 8 years, I have led the transcriptional profiling of a large population-based DCIS cohort, aiming to identify a gene signature predictive of patient benefit from radiation therapy. This has already yielded one paper in collaboration with the Dumeaux lab (Rizvi *et al.*, *Breast Cancer Research* 2025), and four additional manuscripts are nearing completion.

Funding. In 2023 I was awarded a CIHR Project grant as co-PI with E. Rakovitch (Sunnybrook) for $1.96M over four years. With B. E. Hildreth I have submitted several applications to NIH/DoD and the AACR Kidney CancerCAN for renal cell carcinoma work in which I lead the spatial profiling of the tumour microenvironment. T. Sørlie (Oslo) and I, with others, are building a joint ERC proposal for 2027.

EXECUTIVE SUMMARY

Over the past 18 months, *agentic AI* - large language models that no longer merely answer questions but take actions on a researcher's behalf, reading and writing files, running analyses, querying scientific literature, and stringing those actions into plans - has become a practical instrument for scientific research [1]. This WRC program will build **Wigamig**, a shared agentic-AI infrastructure for Western's Bioconvergence Centre. Wigamig lets research groups work independently while pooling agents, data, and analyses when collaboration benefits them. The next term will deliver (i) a reference AI *commons* available to all in Bioconvergence, (ii) working software built on mature agentic AI platforms, (iii) faculty-wide best practices for secure use that respects EDID principles, and (iv) trained personnel prepared to lead AI-enabled biomedical research.

CONTEXT AND OBJECTIVES

Bioconvergence will catalyze interdisciplinary research across medicine, engineering, and science at Western. Its computational backbone must reflect how research at such a centre actually happens: groups form, collaborate, and re-form around shifting scientific questions. To date, three paradigms have dominated AI's entry into biomedicine: *Lab-in-the-Loop (LitL)*, in which AI proposes molecules for contract labs to test [2–4]; *self-driving laboratories*, in which AI and robotics form closed hardware loops [5–8]; and the broader program of *agentic science* [1, 9]. These strategies are best suited to the pharmaceutical industry, which requires a single fixed linear pipeline. Bioconvergence requires an AI paradigm that supports dynamic collaborations for a diverse range of research end-points exploiting a broad range of biotechnologies, while simultaneously protecting privacy, security and group sovereignty.

Our answer to this challenge is Wigamig[1], a novel AI paradigm that re-conceptualizes Bioconvergence as a *village*: a network of individuals, *groups* (a PI and their HQP), *collaborations* (sets of groups with projects), *cores* (e.g. a proteomics facility), and *administration*. Wigamig is similar to Genentech's CLADD [10].

The AI layer. An *AI agent* is an independent AI specialist, each with its own memory, its own perspective on the task at hand, and its own skill set that improves as the researcher interacts with it over time. Wigamig has seven reference agents, each responsible for different tasks: *oracle* (research memory), *bookworm* (literature, databases), *blacksmith* (computation, statistics), *adversary* (methodological audit), *artist* (visualization, education), *conscience* (EDID/bias review), and *lawyer* (patent counsel). The *commons* is the centre-wide AI infrastructure every member draws on: these reference agents, data-governance rules, and baseline workflows. I made a video (see [11] for key moments) showing the individual-researcher experience using Wigamig in biotherapeutic research. On top of the commons, each group or core builds its own *AI toolkit* which can have both private and shared components. Collaborations can also define their own data-scopes and privacy policies which dissipate when the collaboration ends. A centre-wide *administration layer* with a governance body sets rules of conduct, security-and-privacy enforcement agents (the village's "cops"), EDID watchdogs, and a centre-wide oracle that aggregates institutional knowledge across every project over time.

Choreography, not orchestration. Coordinating AI across many groups at a centre presents an architectural challenge addressed by this WRC. Existing agentic AI models use *orchestration*, where a central controller tells every group's agents what to do (a symphony conductor cueing each instrument). Orchestration fails for Bioconvergence since one controller cannot understand all aspects of a project well enough to route every group's work, creating bottlenecks and challenging each group's sovereignty. We will instead develop *choreography*, where each group runs its own documented pattern using shared agents and rules, without centralized control. This decentralized approach allows units to work independently, to limit the reach of failures, and to preserve each group's autonomy. Every AI agent knows its role in the village.

I have built two prototype choreographies so far. The *drug discovery* choreography can be run by any team in Bioconvergence: AI proposes molecules, a wet-lab partner tests them, the AI refines the next proposal. The *clinical cohort analysis* choreography uses agents (e.g. *blacksmith*) to mine clinical and transcriptomic data

[1] *Wigamig* is a working name, pending consultation with Western's Office of Indigenous Initiatives.

for associations and survival characteristics. Each unit will develop their own choreographies.

Why it matters. Research data is the new oil; without stewardship it fragments into per-project silos that die when projects end. Wigamig turns Bioconvergence's collective knowledge into an accumulating asset under strict project-level privacy. It supports the Centre's mandate for interdisciplinary collaboration towards biotherapeutics. It is a necessary first step towards a feasible plan to support advanced biomedical computing at Western. Wigamig is a vehicle to introduce our researchers to real-world use of agentic AI.

Methodology and Feasibility

Wigamig is an *implementation* program, not an AI-methods program. We build on mature production tools (e.g. Claude Code) atop Western/CFI-funded computing infrastructure; the architecture absorbs new, better tools as the AI ecosystem evolves.

The personal-layer prototype already exists. The challenges for the renewal are: (i) harden the commons with audit logs, declarative permissions, and a publish/subscribe channel linking per-project *oracles* to a centre-wide oracle; (ii) co-develop discipline-specific AI toolkits with the Centre's groups and cores; (iii) grow the choreographies beyond the two existing prototypes; and (iv) develop a faculty-wide best-practices reference.

Security, privacy, and IP. The core engineering problem for Wigamig is to keep the right walls around the right people, agents and data everywhere all the time. Our vision is that rules will "travel with the data": every agent reads the policy before acting, and a centre-wide audit trail records every access for review. IP stays at Western: agents run on Western-controlled infrastructure by default, requiring an explicit release step. The *lawyer* agent will route, for example, freedom-to-operate checks through the Research & Innovation Office.

Cost transparency. What will this cost and can Bioconvergence ultimately afford it? An *accountant* agent at the administration layer tracks every API call, GPU-hour, and storage unit (metered at source by a single gateway proxy) and tags events by person, group, and core. The first 24 months should give us an open cost model with defensible estimates of per-project costs for Bioconvergence.

Feasibility rests on three things: completing the group/core, collaboration, and administration layers; a mature tool stack; and engagement from Bioconvergence members. The first two are software-engineering challenges. The third is a "social-engineering" problem, where graduate-level HQP may be the lever.

Training and Mentorship

Wigamig's implementation-first design makes it unusually accessible as a training platform. Because the system is assembled from mature production tools, projects exist for every level of training: undergraduates can build new agents, graduate students can lead a choreography or build a new toolkit tailored to their research interests. Each trainee is also mentored in research data stewardship and responsible-AI practice. Regular AI demonstrations and workshops, an annual Birthing a New Agent hackathon, and a re-vamped AI-user focused BIOCHEM 9552 course will anchor the ecosystem.

Engagement and Knowledge Mobilization

We will work with each group to co-design an AI toolkit for its discipline. Knowledge mobilization takes three forms: open-source release of the commons and reference AI toolkits under a permissive license; an AI best practices white paper for Western; and a branded demonstration deployment that other institutions can mirror. Regular reporting will flow to the appropriate Western offices e.g. security issues to WTS, computing needs to the VPR.

Equity, Diversity, Inclusion, and Decolonization

Deployed uncritically, AI pipelines threaten EDID by amplifying biases in their training data; built carefully, however, they can extend EDID mandates significantly (see my video [11] at approx. 47:14, for examples). The centre-wide *conscience* agent, which reviews experimental design, literature, language, and visualizations for bias, representation gaps, and colonial framings, is itself a first-class research artefact of this program.

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Table 1: Key moments in the Wigamig seminar video.

Time	Topic
03:24	What is agentic AI?
08:57	What does Wigamig look like for anyone in Bioconvergence?
20:00	Demonstration start-up
21:40	The biotherapeutic choreography starts
46:43	The biotherapeutic choreography ends
47:14	The *oracle* (remembering important things)
	The *conscience* (EDID; example)
50:20	Security, privacy
52:00	Similar strategies in pharma